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TOPICAL REVIEW

A Systematic Literature Review of Agentic AI: Definitions, Architectures, and Challenges

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ABSTRACT This Systematic Literature Review (SLR) synthesizes the rapidly evolving state of Agentic AI, analyzing peer-reviewed studies published between 2021 and 2025. We organize the current body of knowledge into a taxonomy comprising five distinct research fields: Memory Cognition, Networking Systems, Trust Safety, Evaluation Limits, and Applications Use-Cases. Our intersection analysis reveals a research landscape currently in an "exploitation" phase, heavily skewed toward Applications Use-Cases (31.9%) and the practical implementation of Autonomy. While the community is successfully operationalizing existing autonomous capabilities, we identify a critical "research desert" regarding Adaptability and New Models. The findings indicate a scarcity of architectural frameworks designed to support long-term resilience and dynamic behavior modification in unforeseen environments. To advance Agentic AI from brittle prototypes to robust real-world systems, we conclude that future research must pivot from solely demonstrating autonomy toward engineering adaptability and developing novel modeling paradigms that prioritize responsiveness to change.

INDEX TERMS Agentic AI, artificial intelligence, systematic literature review, taxonomy.

I. INTRODUCTION

The rapid evolution of Artificial Intelligence (AI) over the past decades has reshaped how intelligent systems are conceptualized, designed, and deployed. Whereas early AI was regarded primarily as a tool for automating narrow tasks under strict human supervision, recent advances have shifted the field toward systems that demonstrate higher levels of independence, adaptability, and goal-directedness [1], [2]. This progression has brought increasing attention to the notion of Agentic AI: systems that do not merely react to inputs, but actively pursue objectives, adapt strategies, and initiate actions in dynamic environments.

The concept of agency has long been central to philosophy, psychology, and the social sciences, where it is broadly understood as the capacity of an entity to act autonomously and intentionally. Translating this notion into AI, however, remains challenging. Some researchers equate Agentic AI with autonomous agents or multi-agent systems, while others interpret it as an emergent property of large-scale learning systems—such as Large Language Models (LLMs)—when embedded within feedback loops, memory structures, or tool-use frameworks [3]. This diversity of perspectives has led to definitional ambiguity, raising the question of whether current AI systems can genuinely be considered "agentic," or whether they simply simulate agency through behavior.

Despite these uncertainties, Agentic AI is gaining momentum across areas such as cognitive architectures, reinforcement learning, LLM-based agents, and human-AI collaboration [4]. Yet, significant challenges remain. Technically, it is unclear how to design architectures that seamlessly integrate perception, planning, reasoning, and memory at

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scale. Methodologically, there is no consensus on how to measure agency in AI or to benchmark proactivity, goal-directedness, and intentional behavior. Ethically, concerns about accountability, anthropomorphism, and trust persist, particularly as AI systems appear more autonomous and human-like in their decision-making.

Given the growing interest and fragmented perspectives, a Systematic Literature Review (SLR) is both timely and necessary. This article presents the results of an SLR conducted according to Kitchenham's methodology [5]. The review process was guided by research questions and executed through a search string applied to major scientific databases. Crucially, this review goes beyond standard synthesis by introducing a novel Intersection Analysis that maps research contributions against identified gaps. This analysis reveals a distinct 'research desert' regarding adaptability, demonstrating that while the field is successfully operationalizing autonomy, it lacks the architectural models necessary for resilience.

The remainder of this article is organized as follows. Section II presents the research methodology, including the research questions, search process, selection criteria, and qualitative evaluation. Section III presents the outcomes obtained from the research and selection process, while Section IV details the theoretical foundations for Agentic AI. Section V discusses the results of the SLR, introduces the proposed taxonomy, addresses the research questions, and analyzes challenges and trends. Finally, Section VI concludes the article with final remarks and directions for future research.

II. SEARCH METHODOLOGY

This SLR follows the research methodology proposed by Kitchenham [5] and aligns with the reporting principles of the PRISMA 2020 statement [6] to ensure transparency and rigor. The methodology establishes the following steps:

1) **Research Questions:** definition of the questions that guide the investigation process;
2) **Search Process:** identification of relevant studies in scientific databases;
3) **Selection Process:** application of criteria to filter the most suitable studies; and
4) **Qualitative Evaluation:** final assessment of the selected studies to ensure alignment with the research objectives.

A. RESEARCH QUESTIONS

The first step consisted of preparing questions to guide the search. One main question (MQ) was formulated, from which four specific questions (SQs) were derived. The MQ aims to map the state of the art of Agentic AI—systems equipped with language models or other components capable of planning, deciding, and executing actions autonomously, with or without human supervision.

MQ: What are the general approaches to Agentic AI research?

The SQs address complementary aspects of the MQ:

- **SQ1:** How does the literature define and delimit "Agentic AI"?
- **SQ2:** What architectures, frameworks, and design standards predominate?
- **SQ3:** Which application domains already demonstrate practical value?
- **SQ4:** What risks, evaluation challenges, and research gaps remain open?

B. SEARCH PROCESS

This step focused on identifying scientific works capable of answering the proposed questions. The search was conducted exclusively in databases widely used in engineering and computing research: ACM Digital Library,[1] Elsevier[2]IEEE Xplore Library,[3] Nature,[4] Science Direct,[5] and Springer Link.[6]

To maximize coverage, a search string was designed and iteratively refined based on preliminary analyses of outcomes.

Title: "agentic ai" OR "AI agent" OR "autonomous agent" AND ("LLM" OR "large language model")

The string was applied to titles and full texts. Its exact implementation varied slightly depending on the syntax required by each database.

C. SELECTION PROCESS

The initial search covered the period from 2021 to mid-2025. Metadata from the retrieved papers were tabulated to create a database[7] used for subsequent analysis and filtering.

Although the search string was designed to maximize precision, many irrelevant results were retrieved (e.g., duplicates, unrelated studies). Therefore, an exclusion process was applied, based on the following exclusion criteria (EC):

- **EC1:** Studies not published in scientific journals or conference proceedings;
- **EC3:** Studies not directly related to Agentic AI.
- **EC3:** Duplicated studies retrieved from multiple databases;

While EC1, EC2, and EC3 were straightforward to apply through metadata queries, the final selection of the studies required a manual screening of the texts, and it was divided into the following two steps:

1) **Filtering by title and abstract:** Initial screening to check whether the study addressed Agentic AI;
2) **Filtering by full text:** In cases where the title and abstract were inconclusive, the full paper was read

[1]https://dl.acm.org/
[2]https://www.elsevier.com/products/journals,
[3]https://ieeexplore.ieee.org/Xplore/home.jsp
[4]https://www.nature.com/nature/research-articles
[5]https://www.sciencedirect.com/
[6]https://link.springer.com/search
[7]This database is available as supplementary material

to decide inclusion or exclusion. To ensure inter-rater reliability, the selection process was performed independently by two researchers. Disagreements regarding inclusion were discussed until a consensus was reached, with a third senior researcher consulted to resolve persistent discrepancies.

D. QUALITATIVE EVALUATION

The final stage was a qualitative evaluation of the remaining studies. According to Kitchenham [5], it is crucial to define qualitative criteria to assess whether each study meaningfully contributes to the corpus. Therefore, we evaluated the selected articles with regard to the purpose of research, contextualization, literature review, related work, methodology, the results obtained, and the conclusion in accordance with objectives and indication of future studies. Hence, this SLR defined the following qualitative evaluation (QE) questions:

- **QE1:** Does the article clearly show the purpose of the research?
- **QE2:** Does the article adequately describe the literature review, background, or context?
- **QE3:** Does the article present the related work with regard to the main contribution?
- **QE4:** Does the article have an architecture proposal or research methodology described?
- **QE5:** Does the article have research results?
- **QE6:** Does the article present a conclusion related to the research objectives?
- **QE7:** Does the article recommended future works, improvements, or further studies?

III. RESEARCH OUTCOMES

This section analyzes the outcomes of the application of the methodology. Figure 1 summarizes these outcomes according to each phase of the selection process of Kitchenham's methodology.

Moreover, Figure 1 also shows that the initial search resulted in a corpus of 125 studies from across four scientific databases researched, where this selection process used the search string presented in section II-B. After performing the impurity removal of studies (EC1 and EC2), the number was reduced to 115. After applying EC3 to remove the duplicated studies, we obtained 109 papers. Finally, after manual selection of the remaining papers filtering by title or abstract and by full text, we obtained the final number of papers from the selection process of 48.

Figure 2 summarizes the selection process based on a search in scientific databases, after applying the exclusion criteria.The results are: ACM (20) database returned more searches, followed by Elsevier (12), IEEE (8), Elsevier (3), Springer (3), Science Direct (3), and Nature (2).

The methodology step following the exclusion criteria application is the quality assessment of the remaining papers. In this phase, the studies should answer quality-related questions raised by the researchers conducting the SLR. Figure 3 shows the outcomes of this phase, presenting the scores for all 48 papers remaining after the quality evaluation.

A. COMPARISON WITH EXISTING SURVEYS

To contextualize our findings, we compared this SLR with recent related surveys in the field. As shown in Table 1, while other works provide valuable overviews of autonomous agents [1] or domain-specific applications like Industry 4.0 [7], this study is distinct in its methodology and analytical depth. Specifically, this is the only review to employ a PRISMA-compliant selection process combined with an Intersection Analysis (mapping contributions vs. gaps) to quantitatively identify the 'Adaptability Desert' in current research.

IV. THEORETICAL FOUNDATIONS

Having established the quantitative landscape of the field, it is essential to ground these metrics in a solid conceptual framework. This section explores the theoretical foundations of Agentic AI, moving from bibliometric statistics to the philosophical and computational definitions that underpin the selected studies.

A. ORIGINS OF THE CONCEPT OF AGENCY

Firstly, it is important to establish the differences between autonomy and agency. Though often used interchangeably, both terms are conceptually distinct. Autonomy refers to operational independence—an agent's ability to function without constant external control. Many autonomous systems, such as robotic vacuum cleaners or cruise control software, perform complex tasks without human input but do so without goals, reasoning, or long-term planning.

In contrast, agency implies goal-directedness and intentional behavior. An agentic system not only acts independently but does so in pursuit of objectives it selects, constructs, or interprets. This distinction is central to debates about responsibility, ethics, and safety in AI systems.

Broadly, the concept of agency refers to the capacity of an entity to act independently and make its own choices. This general notion emphasizes intentionality, decision-making autonomy, and the ability to influence outcomes within a given environment.

In philosophy, the idea of agency has been extensively explored in relation to concepts like intentionality, free will, and moral responsibility. Philosophers such as Daniel Dennett [9] have contributed significantly to contemporary debates, particularly regarding intentional systems and the nature of agenthood. The author introduced the "intentional stance", a framework suggesting that entities exhibiting goal-directed behavior can be interpreted as having beliefs and desires, regardless of their internal mechanisms. Following Gunkel's view [10], moral agency in artificial systems challenges us to rethink ethics not as a human-exclusive domain but as a framework that must also address the actions and responsibilities of autonomous non-human agents.

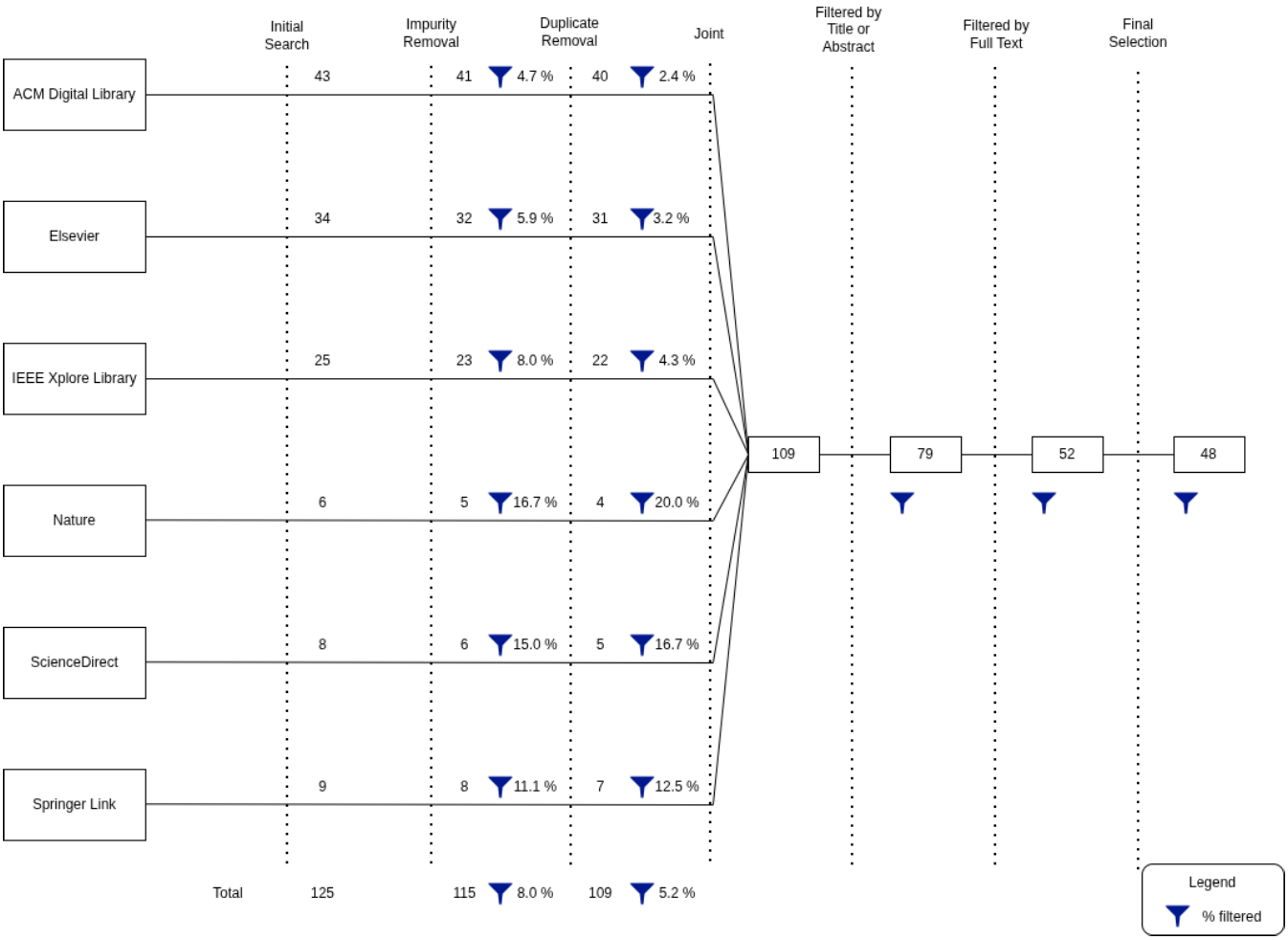

FIGURE 1. Search Process.

TABLE 1. Comparison of this SLR with recent related surveys.

Feature	Acharya et al. [7]	Piccialli et al. [8]	Hosseini & Seilani [9]	This Work (Moralles et al.)
Year	2025	2025	2025	**2026**
Primary Scope	General Agentic AI Overview	Distributed AI for Industry 4.0	Agentic AI for "Smart Futures"	**Cross-Disciplinary Agentic AI**
Methodology	Narrative Survey	Narrative Survey	Systematic Review	**SLR (Kitchenham & PRISMA)**
Taxonomy Focus	General Concepts	Industrial Autonomy Levels	Application Domains	**Unified Agentic Capabilities**
Key Contribution	Comprehensive Definition	Domain-Specific Framework	Trend Analysis	**Intersection Analysis (Heatmap)**
Gap Identification	General Challenges	Connectivity & Standards	Implementation Barriers	**"Adaptability Desert" & Resilience**

In the social sciences, Anthony Giddens [11] positioned agency within the theory of structuration, emphasizing the dynamic interplay between individual action and social structures. For the author, agency implies that individuals possess knowledge and the capacity to intervene in the world, even as they are shaped by broader institutional and cultural forces.

From a psychological perspective, Albert Bandura's theory of human agency [12] introduced a nuanced view of agency as the ability to intentionally influence one's own functioning and life circumstances. The author emphasized self-efficacy and self-reflective thought as core mechanisms enabling human agency, which has been particularly influential in cognitive and behavioral sciences.

Lastly, in cognitive science, the study of agency overlaps with work on intentional action, decision-making processes, and theories of mind, often focusing on how agents (biological or artificial) represent goals and translate them into purposeful actions.

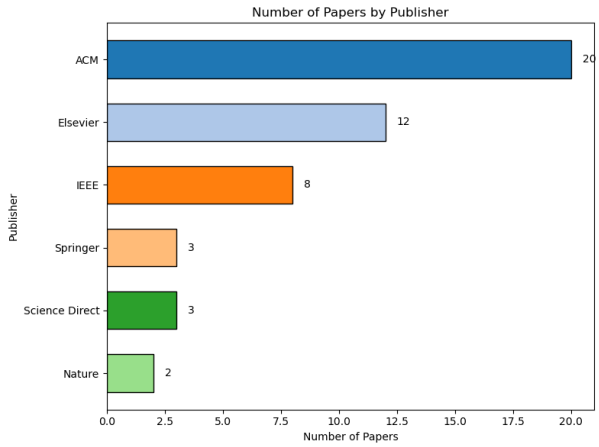

FIGURE 2. Distribution of papers by publishers.

B. AGENCY IN AI

As the field of AI matured, the notion of agency found increasing relevance, particularly in discussions surrounding autonomous systems and intelligent agents. Early AI systems, particularly those associated with Good Old-Fashioned AI (GOFAI), were predominantly symbolic in nature, relying on rule-based reasoning and explicit knowledge representations. These systems, while capable of sophisticated problem-solving, operated primarily in a reactive mode—responding to inputs based on predefined rules rather than proactively pursuing goals.

The transition toward Agentic AI emerged with the development of autonomous agents in the late 20th century. Influential in this shift was the work on Belief-Desire-Intention (BDI) architectures, which sought to formalize decision-making processes in software agents by modeling them as entities possessing mental attitudes analogous to those of humans [13]. These agents were distinguished from traditional expert systems by their ability to generate goals, plan actions, and modify behavior in response to environmental feedback.

Further advancements in reinforcement learning, evolutionary algorithms, and multi-agent systems continued to push AI from passive, task-specific systems toward proactive, adaptive entities. Modern discussions around Agentic AI often focus on systems that not only exhibit autonomous behavior but also demonstrate goal-seeking tendencies, adaptivity to changing environments, and the capacity to formulate plans or sub-goals to achieve overarching objectives.

More recently, the integration of Large Language Models (LLMs) with external tools has given rise to agentic frameworks capable of executing complex, multi-step tasks, reinforcing the shift from tool-like AI to agent-like AI in both research and application domains. This evolution from reactive to agentic systems represents a conceptual transformation in AI research: where early systems were designed to follow instructions, Agentic AI aspires to formulate its own objectives and pursue them independently, often in unpredictable or novel contexts.

C. DEFINING AGENTIC AI

Despite its growing importance in research and industry, Agentic AI remains a conceptual frontier with no universally accepted definition. The term is used in overlapping and sometimes conflicting ways across disciplines, reflecting the ongoing evolution of AI systems and theoretical perspectives on artificial agency.

1) COMPETING AND COMPLEMENTARY DEFINITIONS

A major source of ambiguity arises from the overlapping use of terms such as "autonomous agents," "intelligent agents," and "Agentic AI." While these concepts share similarities, they are not synonymous.

- **Agentic AI as a Synonym for Autonomous Agents:** In some works, Agentic AI is treated as conceptually equivalent to autonomous agents, particularly in the context of multi-agent systems or robotics. The work of [14] defined an agent as "a computer system situated in some environment that is capable of autonomous action in this environment in order to meet its design objectives". Under this view, agency denotes a system's ability to operate independently of direct human intervention, often with decision-making mechanisms explicitly encoded in its architecture (*e.g.*, Belief-Desire-Intention models, reinforcement learning agents).
- **Agentic AI as a Property of Certain AI Systems:** Other perspectives, particularly those concerning Large Language Models (LLMs), adopt a more flexible interpretation. For example, in frameworks such as tool-use agents (e.g., AutoGPT, ReAct agents), agency-like behavior emerges not from symbolic representations but from interaction with external tools and dynamic goal execution. In the view of [15], agentic behavior is an emergent property of the system's capacity to plan, reason, and act in coordination, even without explicit symbolic intentions. This reflects the growing trend of treating LLM-based systems as "agents," especially when paired with control loops or planning modules.

2) CHARACTERISTICS OF AGENTIC AI

Although definitions differ, recent literature converges on four primary characteristics distinguishing Agentic AI from purely reactive or tool-based systems. To address the need for a concrete engineering standard, we propose these characteristics as an Operational Definition for Agentic AI. This framework, detailed in Table 2, provides specific, measurable criteria for system design and benchmarking:

These characteristics provide operational criteria by which AI systems can be classified as agentic in contemporary contexts. However, few systems embody all four simultaneously, making this both an aspiration and an ongoing area of research.

D. THEORETICAL FOUNDATIONS IN AI

Artificial Intelligence has evolved through distinct paradigms, each offering perspectives on modeling intelligent behavior.

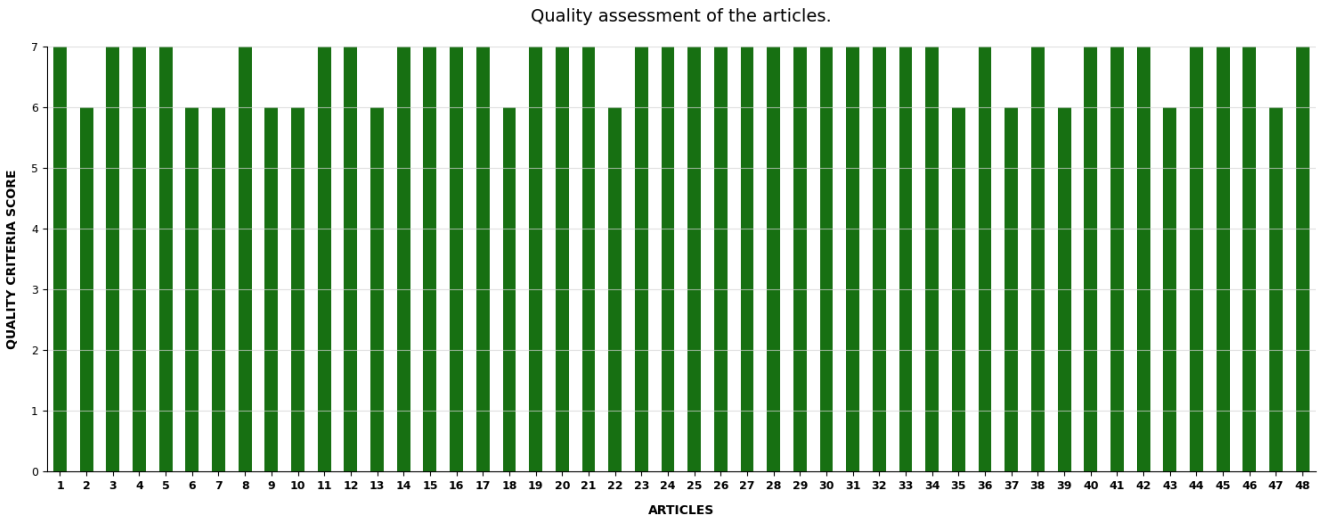

FIGURE 3. Quality assessment of the final selected articles.

TABLE 2. Characteristics of agentic AI.

Characteristic	Description
Autonomy	Makes decisions without constant human oversight.
Goal-Directness	Pursues explicit or emergent goals, whether human-defined or internally generated.
Adaptivity	Learns from and modifies behavior based on interaction with the environment.
Proactivity	Initiates actions to achieve anticipated goals, rather than only reacting to external stimuli.

The foundations of Agentic AI span this evolution, from symbolic systems of the early AI era to modern learning-based and generative architectures supporting increasingly autonomous and adaptive agents.

1) CLASSICAL AI AGENTS

- **Symbolic AI and GOFAI:** Early AI research, often referred to as Good Old-Fashioned Artificial Intelligence (GOFAI), emphasized symbolic reasoning, where intelligence was treated as the manipulation of human-readable symbols. Systems were programmed with explicit rules and knowledge bases, making them transparent but rigid. Expert systems such as MYCIN, designed to diagnose bacterial infections using medical inference rules, and DENDRAL, developed to infer molecular structures from mass spectrometry data, exemplified this approach. While highly effective within their constraints, these systems operated strictly on static knowledge bases provided by human experts. As a result, they performed reasoning in narrowly defined domains but lacked true agency—specifically the capacity for adaptivity, creativity, or generalization beyond their pre-programmed logic.
- **BDI Architectures:** A milestone in formalizing agency was the Belief-Desire-Intention (BDI) architecture [13], inspired by philosophical models of human reasoning. BDI agents maintain internal representations of:
  - Beliefs (information about the environment),
  - Desires (goals to achieve), and
  - Intentions (plans the agent commits to).

  The BDI model bridged symbolic logic and agent-oriented programming, enabling deliberative agents to reason about future states and adapt dynamically. However, BDI agents remained dependent on symbolic representations and often proved brittle in uncertain or unstructured environments.

2) MODERN APPROACHES

- **Multi-Agent Systems (MAS):** Modern Agentic AI frequently develops within MAS frameworks, where multiple agents interact in shared environments [14]. These interactions can involve the following.
  - Coordination (synchronizing actions),
  - Cooperation (working toward shared goals),
  - Negotiation or competition (resolving conflicting interests).

  MAS research emphasizes distributed intelligence, communication protocols, and emerging behaviors from local interactions, with applications in robotic swarms, smart grids, and simulated societies.
- **Reinforcement Learning Agents:** Reinforcement Learning (RL) trains agents via trial-and-error interaction, updating policies to maximize cumulative rewards. Modern deep RL has shown impressive capabilities in dynamic domains (e.g., AlphaGo, autonomous navigation, robotics) [16]. RL agents embody adaptivity and goal-directedness, aligning with Agentic AI. However, they typically lack explicit internal representations of beliefs or intentions, instead focusing on optimization—a key distinction in debates about full-fledged agency.

- **LLM-based Agents:** A recent frontier involves Large Language Models (LLMs) such as GPT-5, PaLM, and Claude. When paired with tools and memory modules, LLMs can reason, plan, and act [15], [17]. Systems like AutoGPT, BabyAGI, and OpenAI's Agents API demonstrate capabilities such as:
  - Task decomposition,
  - Environment exploration,
  - Autonomous tool use,
  - Long-term planning.

  These systems often lack explicit architectures like BDI, yet display emergent agent-like behavior. This raises the question of whether agency requires structured cognition or if it can emerge from flexible, language-driven interfaces.

E. TECHNICAL CHALLENGES

Developing truly agentic systems presents challenges beyond those faced by traditional AI models. While AI has advanced in learning, perception, and reasoning, integrating these into coherent, proactive agents remains a major engineering hurdle.

1) ARCHITECTURAL COMPLEXITY

Designing agentic systems requires combining components such as:

- Perception modules (e.g., sensors, natural language input),
- Memory systems (episodic and semantic memory),
- Planning and reasoning engines (symbolic or neural),
- Learning mechanisms (supervised, reinforcement, or self-supervised),
- Action execution interfaces (e.g., APIs, physical actuators).

Such integration demands modular yet cohesive architectures supporting dynamic goal management and adaptive decision-making. In practice, this often results in fragility, where small changes in one module unpredictably affect overall behavior.

In LLM-based agentic systems (*e.g.*, AutoGPT), reliance on external tools and recursive prompting introduces added complexity, particularly around prompt management, error handling, and context tracking [17].

2) EVALUATION OF AGENTIC CAPABILITIES

Measuring 'agency' is more difficult than assessing task performance. Traditional metrics like accuracy or reward fail to capture agentic behaviors such as:

- Long-term goal pursuit,
- Self-initiated actions,
- Adaptation to novel environments,
- Coordination with humans or other agents.

No standard benchmarks currently exist for quantifying agency. Embodied AI testbeds (e.g., AI Habitat, WebArena) and planning challenges provide partial coverage but lack formal definitions of proactivity or intention [18]. Moreover, stochasticity and opacity in agentic systems—especially LLM-based ones—complicate replication and validation. Without explainability and observability tailored to agentic systems, debugging remains difficult.

3) GENERALIZATION AND ROBUSTNESS

A persistent issue is generalization across tasks and domains. Many agentic systems, while effective in narrow contexts, fail in open-ended settings. Limited robustness compounds this problem: minor input changes can yield unpredictable behavior, especially when reasoning is not causally grounded [19], [16]. These issues are critical in safety-sensitive domains such as healthcare, finance, and autonomous
driving.

Developing agents that can continually learn, transfer knowledge across domains, and maintain stable behavior over time remains an open challenge.

4) COMPUTATIONAL AND RESOURCE DEMANDS

Agentic AI systems, particularly those based on foundation models, are resource-intensive, requiring costly fine-tuning, inference cycles, and memory management. Agents tasked with reasoning, API interaction, or long-term logging demand significant computational and storage resources [20]. This raises concerns about scalability, feasibility, and sustainability, especially in resource-constrained or edge environments.

V. ANSWERS TO THE QUESTIONS AND DISCUSSIONS

With a clear theoretical definition of 'Agency' now established—distinguishing it from simple autonomy, it is possible to effectively evaluate the selected literature against these criteria. This section answers the research questions by synthesizing the 48 selected studies through our proposed taxonomy and intersection analysis.

Towards this goal, Figure 4 shows our proposed taxonomy, which delineates the 48 resulting studies according to 5 fields of research. Table 3 complements the figure by presenting the meaning of each class defined in the proposed taxonomy and classifying each paper within the research fields defined in the taxonomy. Unlike prior surveys that isolate specific sub-domains, such as exclusively analyzing LLM-based agents or Multi-Agent Systems, this taxonomy integrates these historically distinct streams under the unified 'Agentic AI' framework. This holistic classification serves a generative purpose: it acts as the structural foundation for the Intersection Analysis on the studies' contributions and research gaps, enabling us to diagnose cross-disciplinary maturity gaps—specifically the'Adaptability Desert', that remain invisible when these fields are analyzed in isolation.

Using data from the studies and working in line with the taxonomy, the following sections present the answers to the questions proposed in II-A.

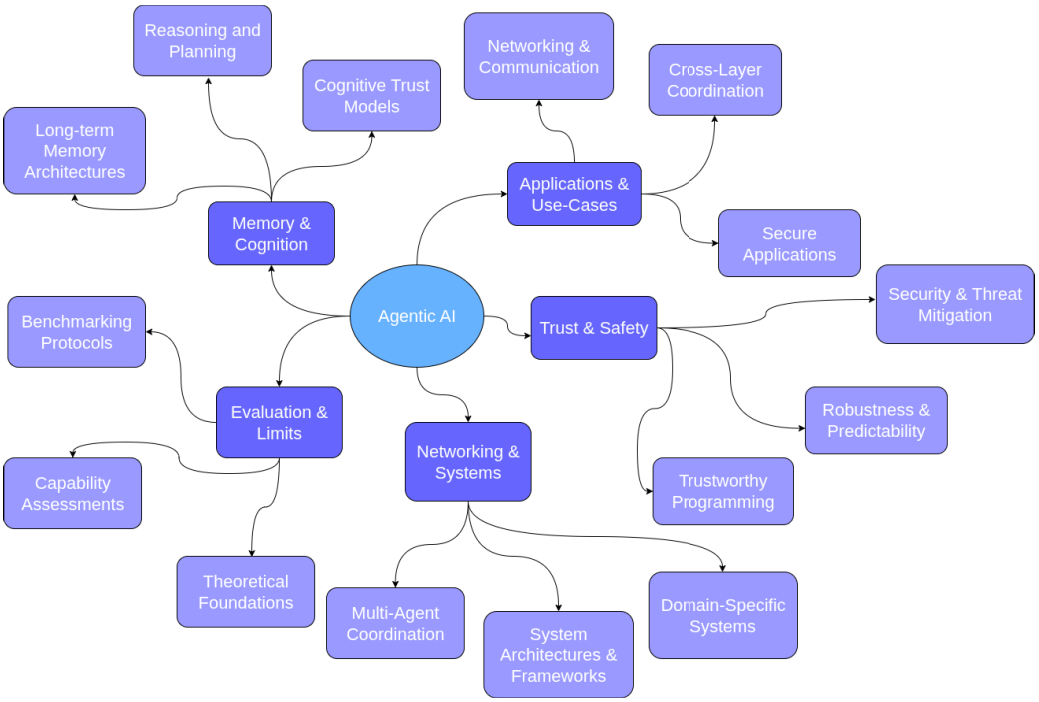

FIGURE 4. Taxonomy of studies.

TABLE 3. Classification of papers within the taxonomy.

Field	Description	References
Memory & Cognition	Focuses on improving agents' reasoning, memory retention, and cognitive architectures for adaptive and context-aware intelligence	[25], [26]
Networking & Systems	Explores distributed frameworks, multi-agent coordination, and system-level designs for scalable and integrated Agentic AI	[8], [27]–[38]
Trust & Safety	Addresses security, robustness, and trustworthiness in Agentic AI, ensuring safe operation in complex environments	[39]–[42]
Evaluation & Limits	Investigates frameworks, benchmarks, and theoretical boundaries for assessing Agentic AI performance and capabilities	[7], [9], [43]–[53]
Applications & Use-Cases	Showcases practical implementations of Agentic AI in domains like networking, industry, and secure applications	[54]–[68]

A. HOW IS THE QUALITATIVE EVALUATION OF AGENTIC AI RESEARCH EVOLVING?

The first question aims to analyze the Agentic AI-related publications quantitatively. Figure 5 presents the outcomes regarding this question.

The distribution of publications over time indicates that the qualitative evaluation of Agentic AI is still in its early stages. From 2021 through much of 2023, only a handful of papers appeared each month, reflecting a period when research was largely concerned with conceptual frameworks and theoretical proposals rather than systematic assessment. This scarcity underscores that evaluation practices had not yet become a central focus within the community.

A sharp spike in publications around 2024 marks a turning point, signaling a surge of scholarly interest in evaluation. This suggests that researchers are now beginning to explore how to test, compare, and validate agentic systems more systematically. However, the immediate decline that

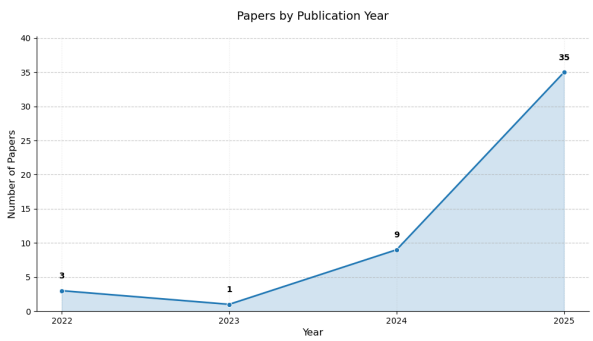

FIGURE 5. Quantity of studies by year.

follows reveals a volatile and still unstandardized evaluation landscape, where practices are emerging but have yet to stabilize. In this sense, qualitative evaluation in Agentic AI is evolving from scarcity to experimentation, pointing to

the need for more consistent benchmarks and reproducible methodologies to consolidate future progress.

B. WHAT COUNTRIES ARE CONDUCTING THOSE RESEARCHES?

The distribution of publications shows that Agentic AI research is being conducted across a diverse set of countries, though with a strong concentration in a few leading regions, as demonstrated in Figure 6.

The United States clearly dominates the field, accounting for the majority of contributions with 21 papers. This is followed by the India and China (5 papers each), and United Kingdom (4 papers), which together form the second tier of contributors. These results suggest that Agentic AI has become a central theme in countries with well-established AI research ecosystems and strong ties between academia, industry, and funding bodies.

Beyond these leaders, only Italy and Australia contribute a modest but noteworthy number of publications (two papers each). Additionally, countries such as Canada, Poland, Iran, Spain, Netherlands, Germany, New Zealand, United Arab Emirates and Bangladesh have emerging contributions, typically represented by single papers. This indicates that while the research community is globally distributed, it remains uneven, with most advances concentrated in a few research hubs and a long tail of countries beginning to engage with Agentic AI.

C. WHAT ARE THE MAJOR AREAS OF STUDIES UNDER RESEARCH?

The goal of this question is to understand which areas of studies are getting the most focus. With this purpose and based on the readings, the 48 papers were categorized according to their areas of knowledge. Table 3 presents each class's descriptions and the related studies' mapping to each class.

Figure 7 shows the percent in which each area is researched. With 31.9% of studies, applications & use-cases is the most searched area, followed by networking & systems and evaluation & limits with both 27.7%, trust & safety with 8.5%, and memory & cognition with 4.3%.

1) MEMORY AND COGNITION

This category groups studies that focus on enhancing agents' reasoning, memory, and cognitive capabilities. Papers here address how Agentic AI can retain long-term conversational memory, support cognitive trust, and perform advance reasoning tasks across multimodal and dynamic environments. Such works explore architectures that allow agents to perceive, interpret, and recall information in ways that mirror human cognitive processes, thereby enabling more autonomous and context-aware decision making.

In [21], the authors designed a human-centric agentic system to help chemists replace hazardous "forever chemicals", with results from experiments that bridged the gap between subject matter experts and open-source discovery tools. Nevertheless, the contributions from this work need expansion to toxicologists and regulators.

The authors in [22] investigated if agents prompted as "entrepreneurs" exhibit human-like entrepreneurial mindsets. Experiments showed that agents displayed coherent mindset profiles, though with stereotype amplification. Future research includes shifting from studying human entrepreneurs to "simulated psychology" of agents.

2) NETWORKING AND SYSTEMS

This category includes papers that examine Agentic AI as part of larger distributed infrastructures, multi-agent systems, and networked environments. Research here often deals with system-level frameworks, orchestration methods, and industry applications ranging from operating systems to collaborative platforms. The emphasis is on scalability, integration, and interoperability – key for deploying Agentic AI in real-world, large-scale systems such as 6G networks, industrial IoT, and financial services.

Regarding user interaction, [23] analyzed the "Gent" chatbot, finding that users often mistake agentic systems for search engines, necessitating greater transparency. In terms of architectural pipelines, [24] used perception agents to infer personality, noting that modular reasoning improved empathy. Similarly, [25] employed cooperative "Planner" and "Retrieval" agents in cardiology to reduce hallucinations and improve retrieval metrics by over 5%, while [26] developed a multi-agent system utilizing 3D Scene Graphs for flexible infrastructure inspection.

In the industrial domain, [27] proposed a framework for autonomous decision-making, and [7] surveyed distributed manufacturing, categorizing autonomy levels toward Industry 6.0. Broader industrial impacts were analyzed in [28], which highlighted decentralized decision-making in supply chains and healthcare, emphasizing the need for governance. Shifting from linear to agentic workflows was explored in [29], demonstrating improved outcomes in code generation.

Specific domain applications include [30] and [31], which reviewed agentic systems in Rheumatology and general diagnostics, respectively, while cautioning on regulatory hurdles. Addressing safety, [32] defined a lifecycle for agentic systems to ensure accountability. Finally, regarding deployment, [33] proposed serverless computing for cost-effectiveness, while [34] introduced "SmartGenie" to automate financial service requests.

3) TRUST AND SAFETY

Trust and safety studies investigate the security, reliability, and ethical dimensions of Agentic AI. These works focus on building resilient agents capable of operating in adversarial or unpredictable environments while maintaining user trust. Topics include robust coordination in multi-agent systems, defending against cyber threats, and embedding mechanisms for transparency and trustworthy programming. This category highlights the growing concern of aligning autonomy with safety guarantees.

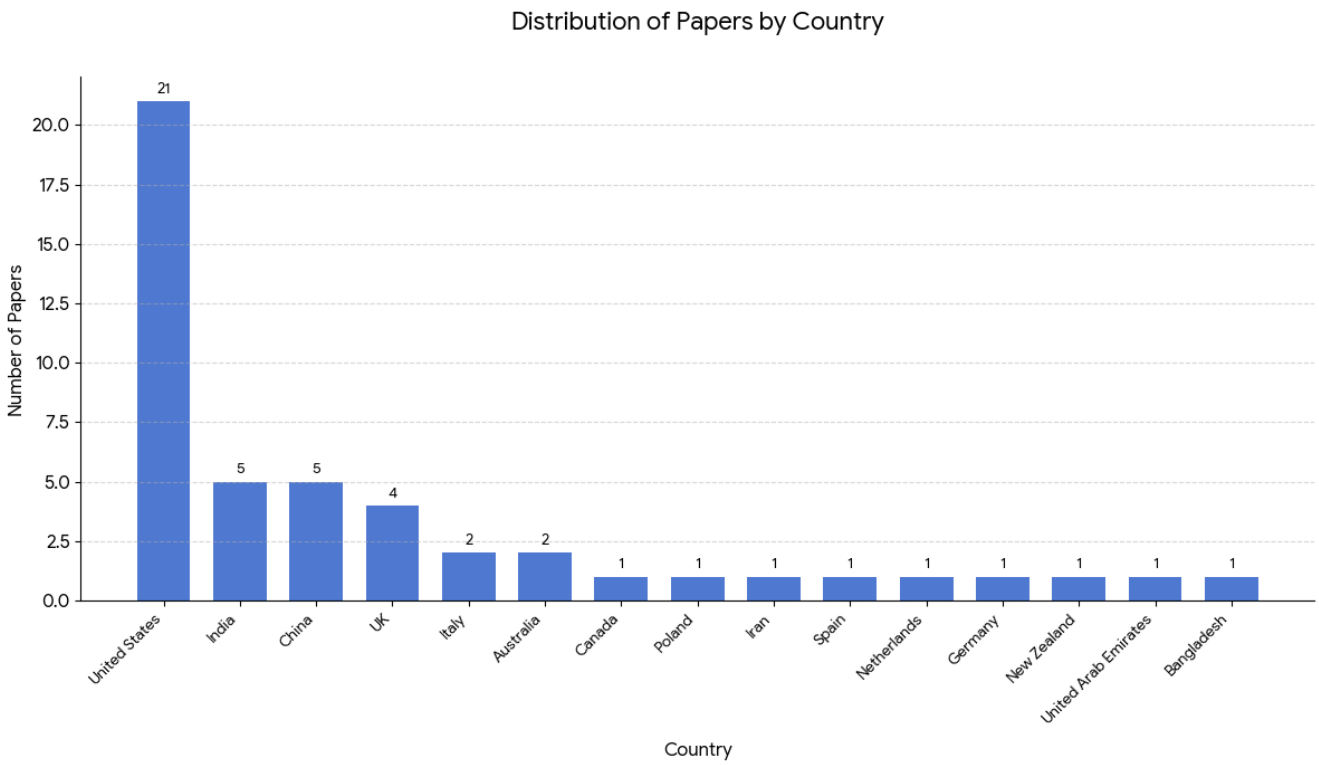

FIGURE 6. Papers by country relation.

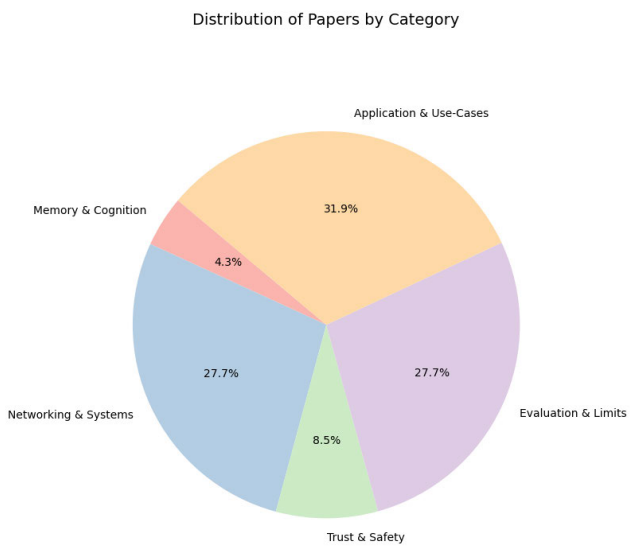

FIGURE 7. Number of studies by area.

Regarding human-agent interaction, [35] found that embodied agents (VR-ECAs) increased user presence and gaze duration through gender matching. However, significant risks regarding social influence remain. In debate simulations, [36] showed that while agents converge on agreements, they are susceptible to logical fallacies. Furthermore, [37] demonstrated that benevolent ageist attitudes in LLMs can reinforce underlying biases in human users, highlighting the need for bias mitigation.

On the operational security front, [38] explored automating Security Operations Centers (SOCs). The study concluded that while automated decision-making is feasible, it introduces new vulnerability risks that demand updated safety frameworks.

4) EVALUATION AND LIMITS

Research in this category addresses methodologies for benchmarking, evaluating and understanding the boundaries of Agentic AI. Papers include frameworks for measuring agent capabilities in planning, translation, or reinforcement learning environments, as well as theoretical explorations of agency limits. By defining metrics and identifying constraints, these works provide critical insights into how Agentic AI can be systematically assessed and compared across different domains.

Regarding frameworks and theory, [40] introduced an eight-dimension classification system to update obsolete evaluation standards, while [43] demonstrated the high sample complexity and difficulty of zero-shot cooperation without prior coordination. Broader implications of these limits and societal integration were discussed in [1] and [44], with the latter outlining a framework for safe autonomy.

In technical systems, [39] found that agents effectively complement—rather than replace—traditional ML in HPC optimization. Similarly, [41] confirmed GPT-4's proficiency in autonomous Linux server management, and [47] highlighted the capacity of agents to minimize downtime in IT operations.

Domain-specific evaluations have yielded significant quantitative metrics. In compliance, [42] showed agents reduced manual GDPR audit efforts by 74%. In finance, [45] reported a 25% improvement in risk model accuracy and a 40% drop in false positives. Additionally, [46] validated the adaptability of vision-based agents using reinforcement learning.

Finally, regarding strategic impact, [48] and [49] analyzed economic benefits in quick commerce and healthcare, respectively. This shifts the focus to deployment models, as reviewed by [8], which characterized the industry transition from assisted "Copilot" tools to fully autonomous "Autopilot" systems.

5) APPLICATIONS AND USE-CASES

This category captures domain-specific implementations of Agentic AI. Studies here present concrete applications such as secure networking frameworks, industry protocols, and cross-layer coordination systems. The focus is on demonstrating the utility of Agentic AI in solving practical problems, showcasing how autonomous agents can be tailored to specialized environments like communications, security, or business processes.

In healthcare and life sciences, agents are increasingly tasked with complex diagnostic and research functions. Reference [61] introduced CARE-AD, a multi-agent framework that predicts Alzheimer's disease risk with superior accuracy by emulating collaborative clinical diagnostics. Reference [54] further demonstrated the efficacy of hybrid agents in oncology, achieving high segmentation performance in breast cancer diagnosis by combining MRI analysis with clinical data. Beyond diagnostics, [60] validated that LLM agents can effectively utilize tools to perform precise clinical calculations, while [52] applied agents to operational data analysis, automating the generation of insights and budget reports. Reference [55] extended these capabilities to biotechnology, utilizing fine-tuned agents to extract knowledge for sustainable microbial protein production.

Industrial and agricultural sectors are similarly leveraging agentic systems for automation and resource management. References [57] and [62] review the role of agents in smart manufacturing and hyper-automation, noting significant productivity gains alongside the need for strategic deployment. Reference [59] provided a concrete implementation of this via the IRCSAD framework, which automates software development for industrial robot control. In agriculture, [64] presented "AgriBuddy," a system assisting farmers with disease detection, while [56] proposed an edge-enabled framework for crop classification on resource-constrained devices, bridging the gap between advanced AI and rural deployment.

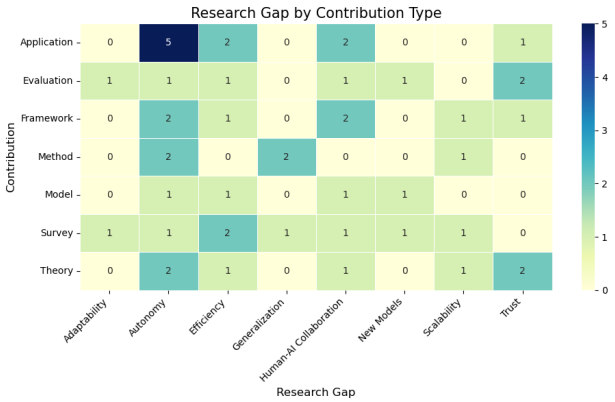

FIGURE 8. Heatmap illustrating the distribution of primary research contributions versus addressed research gaps in the domain of Agentic AI.

Finally, specialized applications in digital services and system architecture highlight the versatility of these agents. Reference [63] introduced "AIcademic," leveraging professor and student agents to enhance active learning in education, while [53] developed "InteRecAgent," which outperforms traditional recommender systems by using LLMs as a central controller. In the realm of security forensics, [58] utilized agents to generate synthetic datasets for mobile investigations. To support these diverse applications, [50] proposed the SCMRAG system to dynamically correct retrieval errors and reduce hallucinations, and [51] formalized "auditability" axioms to ensure these multi-agent systems remain transparent and verifiable in production.

D. ANALYSIS ON THE STUDIES' CONTRIBUTIONS AND RESEARCH GAPS

Furthermore, in order to systematically analyze the related literature, we classified the reviewed works along two complementary dimensions: type of contribution and identified research gap or future vision. Contribution categories capture the main form of added value each study provides, as detailed in Table 4. In contrast, Gap/Future vision categories denote the directions suggested or implied for advancing the field, as shown in Table 5. This classification enables a structured comparison of contributions while simultaneously highlighting where research opportunities remain open and which future trajectories are being prioritized by the community.

To understand the maturity and strategic direction of the Agentic AI field, we conducted an intersection analysis mapping the primary contribution of each reviewed study against the main research gap it sought to address. Figure 8 presents a heatmap visualizing this distribution. The intensity of the color reflects the frequency of publications at each intersection, allowing for immediate identification of research concentrations ("hotspots") and underexplored areas ("deserts").

TABLE 4. Classification of papers according to their main contributions.

Field	Description	References
Survey	Reviews or syntheses that summarize existing knowledge and highlight open issues.	[7], [9], [43]–[53]
Framework	Architectural or platform-based proposals that enable integration or system-level solutions	[25], [26]
Method	Methodological approaches or workflows designed to tackle specific research problems	[8], [27]–[38]
Model	Mathematical, statistical, or computational representations for describing or predicting behavior	[39]–[42]
Evaluation	Empirical studies, experiments, benchmarks, or case analyses validating existing approaches	[39]–[42]
Application	Practical implementations or domain-specific use cases that demonstrate feasibility	[25], [38], [51], [52], [54]–[68]
Theory	Conceptual or formal foundations that establish principles or definitions	[26], [42], [47], [48], [53], [66], [69]

TABLE 5. Classification of papers according to their research gaps/future works.

Field	Description	References
New Models	Reviews or syntheses that summarize existing knowledge and highlight open issues.	[39], [50], [57]
Scalability	Architectural or platform-based proposals that enable integration or system-level solutions	[37], [49], [53], [60]
Autonomy	Capacity of adapting and improving agentic ai approaches to new scenarios and problems	[8], [28], [30], [31], [35], [38], [48], [51], [54], [55], [62], [64], [68], [69]
Generalization	Methodological approaches or workflows designed to tackle specific research problems	[29], [33], [61]
Trust	Mathematical, statistical, or computational representations for describing or predicting behavior	[40]–[42], [44], [56], [66]
Efficiency	Empirical studies, experiments, benchmarks, or case analyses validating existing approaches	[9], [34], [43], [47], [52], [59], [63], [65]
Human-AI Collaboration	Practical implementations or domain-specific use cases that demonstrate feasibility	[25]–[27], [32], [36], [46], [58], [67]

The analysis reveals a field that is currently heavily skewed towards the practical deployment of existing autonomous capabilities, while critical areas required for the next generation of resilient agents remain significantly underdeveloped.

1) THE PREDOMINANCE OF AUTONOMOUS APPLICATIONS

The most immediate finding is the significant concentration of research at the intersection of Application contributions and the Autonomy gap (n=5). This "hotspot" indicates that a substantial portion of current efforts in Agentic AI is dedicated to taking existing concepts of autonomy and applying them to specific domain problems.

This trend is reinforced by the distribution of "Autonomy" across other contribution types. It is the only research gap addressed across nearly all contribution categories—including Frameworks (n=2), Methods (n=2), and Theory (n=2). This suggests that "achieving autonomy" is currently the central, defining characteristic of the field. The research community is actively tackling autonomy from theoretical foundations all the way through to practical implementations.

2) CRITICAL GAPS: THE ADAPTABILITY AND MODELING BOTTLENECK

While the focus on operationalizing autonomy is expected in a growing field, the heatmap highlights a critical oversight in the current research landscape regarding long-term agent resilience.

- **The Scarcity of Adaptability Research:** A defining characteristic of truly "agentic" systems in real-world environments is the ability to adapt to dynamic, unforeseen circumstances. However, our analysis reveals a stark "research desert" surrounding Adaptability. Across the entire corpus, only two studies primarily addressed adaptability, and crucially, neither offered a constructive contribution. There were zero (n=0) Frameworks, Models, Methods, or Applications dedicated to solving the adaptability gap. The existing literature is limited to surveying the problem (n=1) or evaluating existing limitations (n=1). This indicates a significant maturity gap; the field recognizes adaptability is an issue but has not yet moved toward proposing concrete solutions or architectural standards to address it.
- **Stagnation in New Models:** Similarly, the gap for New Models is largely unaddressed. There are zero contributions in terms of Frameworks, Methods, or practical Applications that propose fundamentally new agentic models. The few papers in this area are restricted to surveys or evaluations of existing paradigms. This suggests the field may be reaching a saturation point with current agent architectures, prioritizing the refinement of existing approaches over the development of novel

paradigms necessary to overcome complex hurdles like adaptability or generalization.

The intersection analysis portrays Agentic AI as a field in an ''exploitation'' phase—heavily focused on applying and refining current understandings of autonomous behavior. However, the data reveals a critical lack of ''exploration'' into foundational necessities for robust agents, specifically adaptability and novel modeling paradigms. For Agentic AI to move beyond constrained environments into open-world scenarios, future research must pivot from purely applying autonomy toward developing frameworks and methods that inherently support adaptive behaviors. Prescriptively, to bridge this gap, we recommend that the community: (1) Prioritize 'Resilience Metrics' (e.g., recovery rate from unexpected errors) over standard task accuracy; (2) Move beyond static LLM chains toward 'Metacognitive Architectures' capable of self-modifying their planning logic in real-time; and (3) Establish 'Chaos Engineering' benchmarks for agents to rigorously test adaptability in volatile environments.

VI. CONCLUSION

This systematic literature review synthesized research on Agentic AI between 2021 and 2025, revealing a field that has rapidly transitioned from conceptualization to operational deployment. Contrary to early-stage domains often dominated by theoretical frameworks, our analysis indicates a strong prevalence of Application-driven studies, particularly those focused on achieving immediate Autonomy. This suggests that the community is currently prioritizing the practical implementation of autonomous behaviors over the development of abstract theories.

However, this operational focus has exposed a critical maturity gap in the long-term resilience of these systems. While the discourse is heavily saturated with solutions for Autonomy and Efficiency, there is a stark scarcity of research addressing Adaptability and New Models. The analysis highlights a ''research desert'' where agents are built to function autonomously but lack the architectural foundations to adapt to dynamic or unforeseen environments.

Consequently, the field stands at a pivotal juncture. The current trajectory—optimizing existing models for specific autonomous tasks—risks creating brittle systems that fail outside controlled parameters. To advance Agentic AI from functional prototypes to resilient real-world systems, future research must pivot from purely demonstrating autonomy to engineering adaptability. Bridging this gap requires a shift toward novel modeling paradigms that prioritize how agents evolve and respond to change, rather than solely how they execute pre-defined tasks. To guide this transition, we propose the following concrete research agenda: (1) Architectural Principles: Researchers should move beyond static execution chains toward 'Metacognitive Architectures' that feature distinct modules for self-reflection and real-time plan modification; (2) Evaluation Strategies: Benchmarking must shift from static accuracy metrics to 'Resilience Benchmarks' (e.g., Chaos Engineering for AI) that measure an agent's recovery rate from induced environmental errors; and (3) Research Directions: Future efforts should prioritize 'Adaptive Learning' frameworks that allow agents to update their knowledge base post-deployment without requiring full model retraining.''

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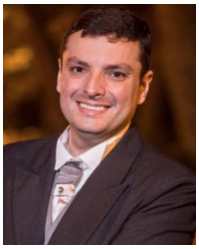

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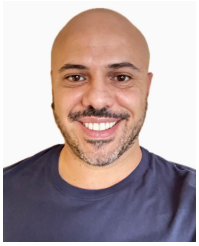

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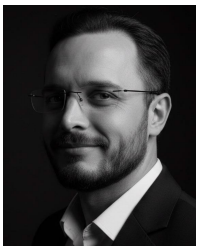

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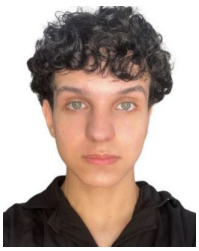

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